Metals and Non-metals

Classification :

Elements are classified into metals, non-metals, and metalloids based on their properties, which are correlated with their placement in the periodic table.

Metals :

There are many substances which are known as metals because some of their unique properties.

For example:- iron, copper, gold, silver, sodium, aluminium, etc.

Non-metal :

There are many elements which are called non-metals because they have some unique properties.

For example:- carbon, sulphur, iodine, oxygen, etc. These properties are generally opposite to metals.

Metalloids :

Metalloids have properties intermediate between the metals and nonmetals.

For example:- Silicon, Boron, Arsenic, Germanium etc.

Physical properties of Metals and Non-Metals :-

PROPERTY	METALS	NON-METALS
1. Luster	Metals have shining surface.	They do not have shining surface. • Except Iodine.
2. Hardness	They are generally hard. • Except Sodium, Lithium and Potassium which are soft and can be cut with knife.	Generally soft. • Except Diamond, a form of carbon which is the hardest natural substance.
3. State	Exist as solids. • Except Mercury.	Exist as solids or gaseous. • Except Bromine.

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4. Malleability	Metals can be beaten into thin sheets. • Gold and Silver are the most malleable metals.	Non-metals are non-malleable
5. Ductility	Metals can be drawn into thin wires.	They are non-ductile.
6. Conductor of heat & electric	Metals are good conductors of heat and electricity. • Silver (Ag) and Copper (Cu): Best conductors of heat. • Lead (Pb), Mercury (Hg) poor conductor of heat.	Non-metals are poor conductor of heat and electricity. • Except Graphite.
7. Density	Generally have high density and high melting point. • Except Sodium and Potassium.	Have low density and low melting point.
8. Sonorous	Metals produce a sound on striking hard surface.	They are not sonorous.
9. Oxides	Metallic oxides are basic in nature.	Non-metallic oxides are acidic in nature.

Chemical Properties of Metals

(A) Reaction of Metal with Oxygen?

Almost all metals combine with oxygen to form metal oxides. When metals are burnt in air, respective metal oxides are formed. Most of the metals show this chemical property

Metal + Oxygen → Metal Oxide

Example:

1. When sodium is heated in air, it starts burning with orange flame and forms a mixture of sodium oxide and sodium peroxide.

$$\underset{\text{sodium}}{4Na} + \underset{\text{Oxygen}}{O_2} \rightarrow \underset{\text{sodium oxide}}{2Na_2O}$$

$$\underset{\text{sodium}}{2Na} + \underset{\text{Oxygen}}{O_2} \rightarrow \underset{\text{sodium peroxide}}{Na_2O_2}$$

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2. When copper is heated in air, it combines with oxygen to form copper(II) oxide, a black oxide.

$$\underset{\text{(Copper)}}{2Cu} + \underset{\text{(Oxygen)}}{O_2} \rightarrow \underset{\text{(Copper(II) oxide)}}{2CuO}$$

3. When aluminium metal is heated or burnt in air, it forms aluminium oxide.

$$\underset{\text{(Aluminium)}}{4Al} + 3O_2 \rightarrow \underset{\text{(Aluminium oxide)}}{2Al_2O_3}$$

4. When lithium is heated or burnt in air, it starts burning with red-tinted flame and gives lithium oxide.

$$\underset{\text{(Lithium)}}{4Li} + \underset{\text{(Oxygen)}}{O_2} \rightarrow \underset{\text{(Lithium oxide)}}{2Li_2O}$$

5. When potassium is burnt in air, it gives potassium peroxide and potassium super oxide.

$$\underset{\text{Potassium}}{2K} + \underset{\text{Oxygen}}{O_2} \rightarrow \underset{\text{Potassium peroxide}}{2K_2O_2}$$

$$\underset{\text{Potassium}}{K} + \underset{\text{Oxygen}}{O_2} \rightarrow \underset{\text{Potassium super oxide}}{KO_2}$$

6. What happens when Silver and Gold is burnt in air :-

Silver and gold do not burn in air, as these are very less reactive metals.

NOTE :- Storage of Sodium, Potassium and Lithium

Since, sodium, potassium and lithium reacts vigorously with oxygen, these metals are kept immersed in **kerosene oil** to prevent any accidental reaction with oxygen or water.

(B) Reaction of Metal oxides with acid

Metal oxides are basic in nature. Thus, when metal oxides react with dilute acid, give respective salt and water.

$$\boxed{\text{Metal oxide} + \text{Acid} \rightarrow \text{Respective salt} + \text{Water}}$$

Example:

1. When sodium oxide reacts with hydrochloric acid, it gives sodium chloride and water.

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$$\underset{\text{Sodium oxide}}{Na_2O} + \underset{\text{hydrochloric acid}}{2HCl} \rightarrow \underset{\text{Sodium chloride}}{2NaCl} + \underset{\text{water}}{H_2O}$$

2. When potassium oxide reacts with hydrochloric acid, it gives potassium chloride and water.

$$\underset{\text{potassium oxide}}{K_2O} + \underset{\text{hydrochloric acid}}{2HCl} \rightarrow \underset{\text{potassium chloride}}{2KCl} + \underset{\text{water}}{H_2O}$$

3. When copper oxide reacts with hydrochloric acid, it gives copper chloride and water.

$$\underset{\text{copper oxide}}{CuO} + \underset{\text{hydrochloric acid}}{2HCl} \rightarrow \underset{\text{copper chloride}}{CuCl_2} + \underset{\text{water}}{H_2O}$$

NOTE :- Amphoteric Metal Oxides

Amphoteric Oxides : some metal oxides, such as aluminium oxide, zinc oxide, etc., show both acidic as well as basic behaviour. Such metal oxides which react with both acids as well as bases to produce salts and water are known as amphoteric oxides.

Example : aluminium oxide and zinc oxide are amphoteric oxides.

(C) Reaction of Metals with water

Metals form respective metal oxide and hydrogen when react with water.

$$\boxed{\text{Metal} + \text{Water} \rightarrow \text{Metal Oxide} + \text{Hydrogen}}$$

Example:

1. When sodium metal reacts with water, it gives sodium hydroxide and hydrogen gas. Sodium metal reacts so vigorously with water that hydrogen gas produced in this reaction catches fire immediately.

$$\underset{\text{Sodium}}{2Na(s)} + \underset{\text{water}}{2H_2O(l)} \rightarrow \underset{\text{sodium hydroxide}}{2NaOH(aq)} + \underset{\text{hydrogen gas}}{H_2(g)}$$

- Reaction of sodium with water is highly exothermic.

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2. Potassium metal reacts vigorously with water and produces potassium hydroxide and hydrogen.

$$\underset{\text{Potassium}}{2K(s)} + \underset{\text{water}}{2H_2O(l)} \rightarrow \underset{\text{Potassium hydroxide}}{2KOH(aq)} + \underset{\text{hydrogen gas}}{H_2(g)}$$

- Reaction of potassium and water is highly exothermic. In this reaction also evolved hydrogen gas catches fire immediately.

3. When aluminium metal reacts with steam, alunimium oxide and hydrogen gas are formed.

$$\underset{\text{aluminium}}{2Al(s)} + \underset{\text{steam}}{3H_2O(g)} \rightarrow \underset{\text{alunimium oxide}}{Al_2O_3(s)} + \underset{\text{hydrogen gas}}{3H_2(g)}$$

- Aluminium does not react with cold or hot water. But aluminium reacts with steam.

4. Iron also does not react with cold or hot water, but when reacts with steam, iron oxide and hydrogen gas are formed.

$$\underset{\text{Iron}}{2Fe(s)} + \underset{\text{Steam}}{4H_2O(g)} \rightarrow \underset{\text{Iron oxide}}{Fe_3O_4(s)} + \underset{\text{hydrogen gas}}{4H_2(g)}$$

5. Reaction of lead, copper, silver and gold with water :-

Metals such as lead, copper, silver and gold **do not** react either with cold water, hot water or steam at all.

Rusting of Iron

Iron reacts with cold water or moisture present in atmosphere very slowly and iron oxide is formed in the form of layer over iron. Iron oxide is red in colour. This process of formation of iron oxide in the form of flakes over the surface of iron is called rusting of iron.

(D) Reaction of Metal Oxides with Water

Most of the metal oxides are insoluble in water, but oxides of alkali metals and alkaline earth metals are soluble in water. Water soluble metal oxides react with water and gives, respective base(Metal hydroxide) . These bases are very strong and are called alkali.

$$\boxed{\text{Metal oxide} + \text{Water} \rightarrow \text{Metal hydroxide}}$$

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Example:

1. When sodium oxide reacts or dissolves in water, it gives sodium hydroxide.

$$\underset{\text{sodium oxide}}{Na_2O} + H_2O \rightarrow \underset{\text{sodium hydroxide}}{2NaOH}$$

2. When potassium oxide reacts with water or dissolves in water, it gives potassium hydroxide.

$$\underset{\text{potassium oxide}}{K_2O} + H_2O \rightarrow \underset{\text{sodium hydroxide}}{2KOH}$$

(E) Reaction of metals with acids

When metal reacts with dilute acid, respective salt and hydrogen gas are formed.

$$\text{Metal} + \text{Dilute Acid} \rightarrow \text{Salt} + \text{Hydrogen}$$

But all metals do not react in the same manner and not produce salt and hydrogen gas. For example copper does not react with dilute acid.

Example:-

1. Sodium metal gives sodium chloride and hydrogen gas when reacts with dilute hydrochloric acid.

$$\underset{\text{Sodium}}{Na} + \underset{\text{hydrochloric acid}}{HCl} \rightarrow \underset{\text{Sodium chloride}}{NaCl} + \underset{\text{hydrogen}}{H_2}$$

2. When sodium metal reacts with dilute sulphuric acid, sodium sulphate and hydrogen gas are formed.

$$\underset{\text{Sodium}}{Na} + \underset{\text{hydrochloric acid}}{H_2SO_4} \rightarrow \underset{\text{Sodium chloride}}{Na_2SO_4} + \underset{\text{hydrogen}}{H_2}$$

3. Reaction of metals with nitric acid

Metals react with nitric acid with another way. Nitric acid which is a strong oxidizing agent, oxidizes hydrogen produced in reaction to water and nitric acid itself is reduced to any of the nitrogen oxide [N_2O (Nitrous oxide), NO (Nitric oxide) or NO_2 (Nitrogen dioxide)].

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Whereas magnesium and manganese reacts with dilute nitric acid and gives respective salt and hydrogen gas.

Magnesium produces magnesium nitrate and hydrogen gas when reacts with very dilute nitric acid.

$$\underset{\text{magnesium}}{Mg} + \underset{\text{nitric acid}}{2HNO_3} \rightarrow \underset{\text{magnesium nitrate}}{Mg(NO_3)_2} + \underset{\text{hydrogen}}{H_2}$$

NOTE:-

1. Zinc and aluminium do not react with concentrated nitric acid, because while reaction a thick layer of their oxides is built which is hard to dissolve. This layer of oxide stops the further reaction.

2. Aqua regia: is a freshly prepared mixture of concentrated hydrochloric acid and concentrated nitric acid in the ratio of **3 : 1**

It can dissolve gold, even though neither of these acids can do so alone. Aqua regia is a highly corrosive, fuming liquid. It is one of the few reagents that is able to dissolve gold and platinum.

(F) Reaction of metals with solution of other metal salts

When more reactive metal react with solution of less reactive metal salt, more reactive metal displace less reactive metal from its salt.

$$M_1 + M_2B(aq) \rightarrow M_1B(aq) + M_2$$

Where M_1 is more reactive metal than M_2 and M_2B is salt of metal M_2.

Such reaction is known as displacement reaction or single displacement reaction.

Example:

1. When iron nail reacts with the solution of copper sulphate, iron displaces copper and forms iron sulphate.

$$\underset{Iron}{Fe(s)} + \underset{Copper\ sulphate}{CuSO_4(aq)} \longrightarrow \underset{Ferrous\ sulphate}{FeSO_4(aq)} + \underset{Copper}{Cu(s)}$$

In this reaction iron is more reactive than copper.

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2. When zinc reacts with the solution of copper sulphate, zinc sulphate and copper metal is formed.

$$\underset{\text{Zinc}}{Zn(s)} + \underset{\text{Copper sulphate}}{CuSO_4(aq)} \rightarrow \underset{\text{Zinc sulphate}}{ZnSO_4(aq)} + \underset{\text{Copper}}{Cu(s)}$$

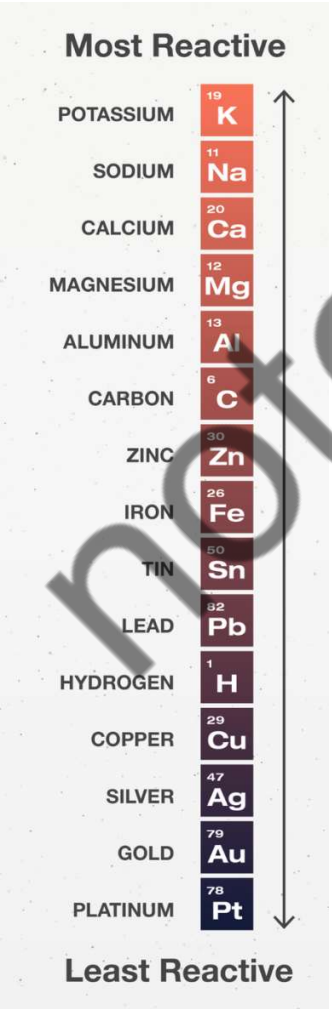
In this reaction, zinc is more reactive than copper, thus zinc displaces copper from copper sulphate.

3. When silver metal is kept dipped in the solution of copper sulphate, no reaction takes place.

Silver metal + copper sulphate solution → No reaction

Reactivity Series

Reactivity series is a list of metals with decreasing order of their reactivity. In this list most reactive metal is kept at the top and least reactive metal is kept in the bottom.



Trick to remember reactivity series

please ———— pottasium

send ———— sodium

cats ———— calcium

monkeys ———— magnesium

and ———— aluminum

zebras ———— zinc

in ———— iron

their ———— tin

lovely ———— lead

happy ———— [hydrogen]

cages ———— copper

made of ———— mercury

silver & ———— silver

gold ———— gold

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Most reactive metal reacts vigorously while least reactive metal do not react with normal condition. For example potassium (K) is kept in the top in the list followed by sodium (Na), calcium (Ca), Magnesium (Mg) and so on. Gold, silver, mercury and copper are kept in the bottom, these metals do not react in normal conditions and hence are called noble metals. This is one of the causes, that gold and silver metals are used in making jewelries.

Chemical properties of Non-metals

(A) Reaction of non-metal with oxygen

Most of the non-metals form their oxides when react with oxygen.

Non metal + Oxygen → Non-metal oxide

Oxides of non metals are acidic in nature. Non-metal oxides forms acid when dissolved in water.

Example:

1. Carbon does not react with oxygen at room temperature. But, when carbon is burnt in the presence of air, it gives carbon dioxide.

$$\underset{\text{carbon}}{C(s)} + O_2 \rightarrow \underset{\text{carbon dioxide}}{CO_2}$$

2. Sulphur does not react with oxygen at room temperature. But, when sulphur is burnt in the presence of air, it gives sulphur dioxide.

$$\underset{\text{Sulphur}}{S(s)} + O_2 \rightarrow \underset{\text{Sulphur dioxide}}{SO_2}$$

(B) Chemical properties of non-metal oxides

Non-metal oxides are acidic in nature. Non-metal oxides turn moist blue litmus paper red. When Non-metal oxides are dissolved in water, it gives respective acid.

Example

1. When carbon dioxide is dissolved in water, it gives carbonic acid. Carbonic acid is a weak acid.

$$\underset{\text{carbon dioxide}}{CO_2(s)} + \underset{\text{water}}{H_2O} \rightarrow \underset{\text{Carbonic acid}}{H_2CO_3}$$

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2. When this sulphur trioxide is dissolved in water, it forms sulphuric acid.

$$\underset{\text{sulphur trioxide}}{SO_3\,(g)} + \underset{\text{water}}{H_2O} \rightarrow \underset{\text{sulphuric acid}}{H_2SO_4}$$

Acid Rain

- Falling of acid in the form of rain is called acid rain. Factories and vehicles emit carbon dioxide, sulphur dioxide and many other harmful gases. These harmful gases go in air and remain present in air.
- Carbonic acid or sulphuric acid falls on earth in the form of rain with rain water and is known as acid rain.
- Acid rain is very harmful for monuments like Taj Mahal, which is made of marble. Acid rain makes water in river, pond, etc. acidic which kills the aquatic animals.

Reaction of metals and non-metals

When a metal reacts with a non-metal few changes takes place with the compound or molecules undergoing in the reaction.

When metal reacts with non-metal following changes occurs.

1. Transfer of electrons

When metals react with non-metals, electrons are transferred from the metal atoms to the non-metal atoms, forming ions. The resulting compound is called an **ionic compound.**

2. Formation of ions

The metal atoms give electrons to the non-metal atoms. The metal atoms become **positive ions (Cation)** and the non-metal atoms become **negative ions(Anaion)** .

To understand these both cation and anaion, we have to understand electronic configuration of elements and their **valencies.**

Valency : Valency is the combining capacity and number of valence electrons present in the outer most shells of an atom is known as valency. Valency may have **negative or positive.**

Ex.

1. In HCl, the valency of hydrogen ion is +1 and the valency of chloride is – 1 .
2. In NaCl, the valency of sodium ion is +1 and the valency of chloride is – 1.

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Valence Electron :- Number of electrons present in outermost orbit of an atom is called valence electrons.

Ex.

1. Electronic configuration of Sodium (Na) is = 2 8 1

There are three shells in sodium atom and the outer most shell has 1 electron can be shared, so valence electron of sodium is 1.

NOTE : -

1. If outer most shell has 1, 2, 3 or 4 electrons these can be **given in sharing** of electrons. so 1, 2, 3, and 4 will be valance electrons.

2. If outer most shell has 5, 6 or 7 electrons these can not be given in sharing of electrons as These **need electrons** to complete their octet.

- Required valance electrons for outer most shell having 5 electrons = 8 - 5 = 3
- Required valance electrons for outer most shell having 6 electrons = 8 - 6 = 2
- Required valance electron for outer most shell having 7 electrons = 8 - 7 = 1

Type of element	Element	Atomic Number	Number of electron in shells			
			K	L	M	N
Noble gases	Helium (He)	2	2			
	Neon (Ne)	10	2	8		
	Argon (Ar)	18	2	8	8	
Metals	Sodium (Na)	11	2	8	1	
	Magnesium	12	2	8	2	
	(Mg)	13	2	8	3	
	Aluminium (Al)	19	2	8	8	1
	Potassium (K)	20	2	8	8	2
	Calcium (Ca)					
Non- metals	Nitrogen (N)	7	2	5		
	Ogygen (O)	8	2	6		
	Flurine (F)	9	2	7		
	Phosphorus (P)	15	2	8	5	
	Sulpher (S)	16	2	8	6	
	Chlorine (Cl)	17	2	8	7	

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Noble Gas

Helium, Neon, Argon, Krypton, Xenon and Radon are noble gas. These elements are called noble because they do not react. In other word, noble gases are non-reactive.

Valence Electron of Noble gas

Valence electron of noble gas is zero. This means outermost orbit of noble gas are completely filled. Helium has total number of 2 (two) electrons in its outermost orbit. And rest of the noble gases has total number of 8 (eight) electrons in their outermost orbit.

Stable and unstable configuration

Completely filled outermost orbit is considered as stable configuration. So, in order to achieve stability every atom has tendency to fill their outermost orbit completely. In other words, since outermost orbit of noble gases are completely filled, so atom of every elements have achieve the electronic configuration of their nearest noble gas.

Formation of chemical compound to achieve stable configuration

In order to achieve stable configuration, an atom forms bond with atom of same element or atom of different elements. Such bonds are called **Chemical Bonds**.

Ionic Bond or Electrovalent Bond

Chemical bond formed after transfer of electrons by two atoms is called Ionic bond or electrovalent bond. In other words, since such type of bond is formed by ions so it is called IONIC BOND. Or this type of bond is formed by electrically charged particles so it is called ELECTROVALENT BOND.

When a metal reacts with a non-metal, bond formed by them is called Ionic Bond or Electrovalent bond. After reaction atoms of metal and non metal get bonded together by electrostatic force of attraction which is called chemical bond.

Example

1.Formation of NaCl (Sodium Chloride)

Atomic number of Na (Sodium) which is a metal = 11

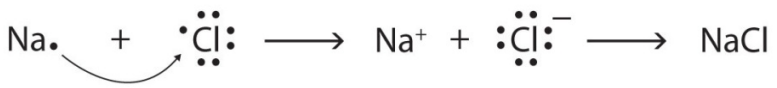
Electronic configuration of Na (Sodium) = 2, 8, 1

Valence electron in sodium atom = 1

Atomic number of Chlorine (Cl) which is a non-metal = 17

Electronic configuration of chlorine = 2, 8, 7

Valence electrons of chlorine = 7



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Properties of Ionic Compounds

(i) Physical nature : Ionic compounds are solids and hard because of the strong force of attraction between the positive and negative ions. These compounds are generally brittle and break into pieces when pressure is applied.

(ii) Melting and Boiling points : Ionic compounds have high melting and boiling points (see Table 3.4). This is because a considerable amount of energy is required to break the strong inter-ionic attraction.

(iii) Solubility : Electrovalent compounds are generally soluble in water and insoluble in solvents such as kerosene, petrol, etc.

(iv) Conduction of Electricity : The conduction of electricity through a solution involves the movement of charged particles. A solution of an ionic compound in water contains ions, which move to the opposite electrodes when electricity is passed through the solution.

NOTE :- Ionic compounds in the solid state do not conduct electricity because movement of ions in the solid is not possible due to their rigid structure. But ionic compounds conduct in the molten state. This is possible in the molten state since the elecrostatic forces of attraction between the oppositely charged ions are overcome due to the heat. Thus, the ions move freely and conduct electricity.

Occurrence of Metal

Metal are found in earth's crust. Metals are also found in seawater in the form of many soluble salts. For example sodium chloride, magnesium chloride, etc. are salts which are found dissolved in sea water.

Minerals :- Elements or compounds which occur naturally in earth crust are called Minerals.

Ores :- Minerals from which metals can be extracted profitably are called ores.

Gangue :- Ores mined from the earth are usually contaminated with large amounts of impurities such as soil, sand, etc., are called gangue.

Extraction of Metals :- Metals are extracted from their ores on the basis of reactivity series. To extract them from ores, metals are divided into following three categories.

(i) Highly reactive : Such as K, Na, Ca, Mg, Al like metals are extracted using Electrolysis.
(ii) Middle reactive : Such as Zn, Fe, Pb and Cu are extracted by Reduction using carbon.
(iii) Less reactive : Ag and Au are found in native state.

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K	(Potassium)	High reactivity
Na	(Sodium)	
Ca	(Calcium)	
Mg	(Magnesium)	
Al	(Aluminium)	
Zn	(Zinc)	Medium reactivity
Fe	(Iron)	
Pb	(Lead)	
Cu	(Copper)	
Ag	(Silver)	Low reactivity
Au	(Gold)	

Metals in the bottom of Reactivity Series

Metals found at the bottom of reactivity series are very less reactive or non-reactive. These metals found in free state in the earth's crust.

Example: Silver and Gold are at the bottom of reactivity series. So silver and gold are found in earth's crust in free state.

Metals in the middle of Reactivity Series

Metals found in the middle of reactivity series are moderately reactive, so they are mainly found in the form of their oxides, sulphides or carbonate. Zinc, Iron, lead and copper fall in the middle of the reactivity series.

Metals in the top of Reactivity Series

Metals lies in the top of the reactivity series are very much reactive. They never find in nature in free state. Potassium, sodium, calcium, magnesium and aluminium fall in the top of the reactivity series.

Metallurgy

Process of extraction of pure metal from its ore is called Metallurgy.

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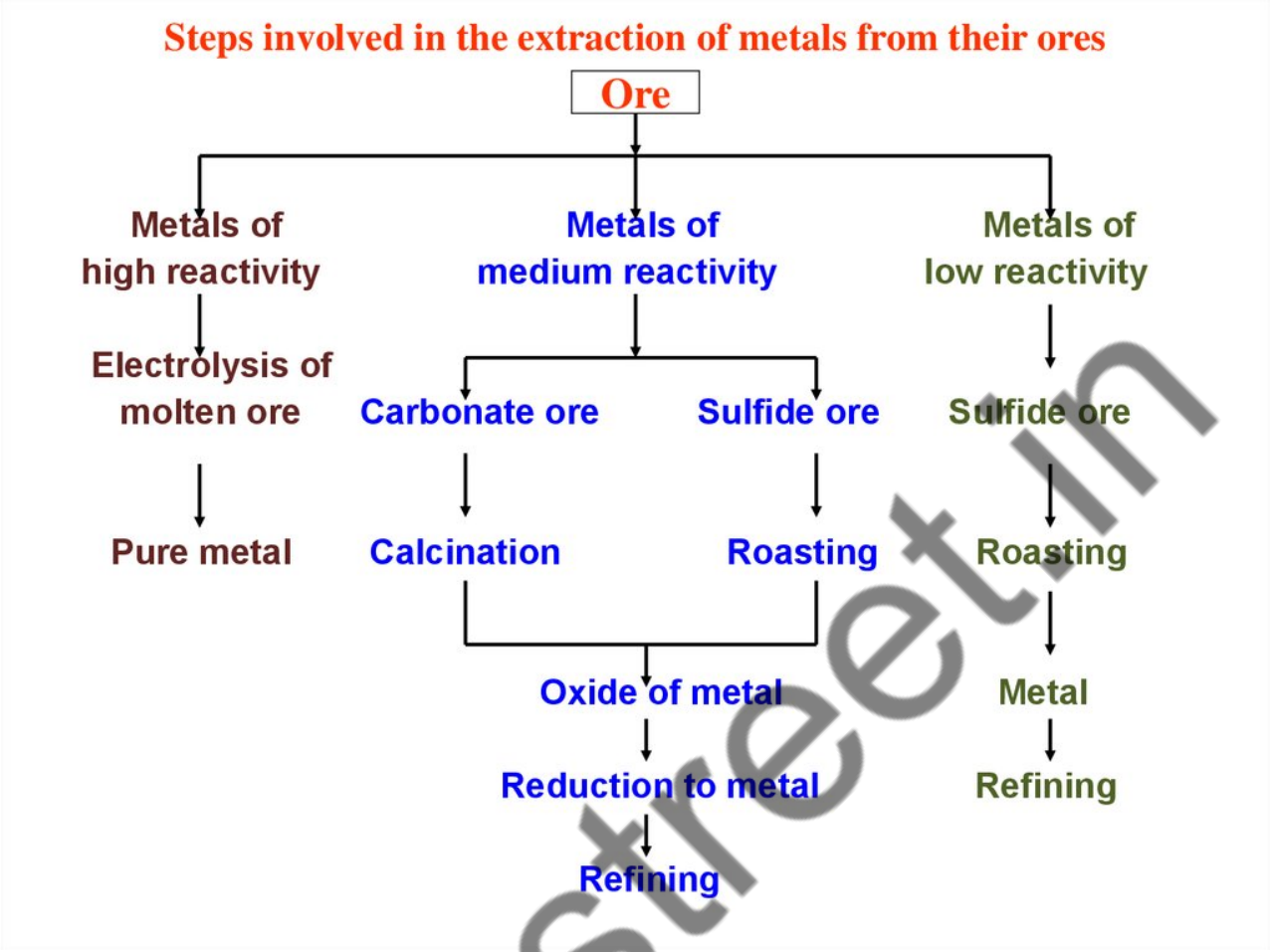

Concentration of Ores

Gaunge : Ores mined usually contained many impurities , such as soil, sand, etc. This impurities of ores is called Gaunge.

First impurities removed from gaunge. Removing of impurities from Gaunge is called **Concentration of Ores** . Concentration of ores is known as enrichment of ores also.

Roasting :- Heating of ores of metals found in the form of sulphide in the presence of air is called the process of Roasting. Roasting converts ores in the form of sulphide to the respective oxides.

Calcination :- Heating of ores of metals found in the form of carbonate in the presence of limited air is called the process of Calcination. Calcination of carbonate ores converts them into respective oxides.

Reduction:- The process of converting metal oxides to respective metals by heating is called Reduction. Reduction converts oxides of metals into respective metals.

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(i) Extracting Metals Low in the Activity Series:

Metals low in the activity series are very unreactive. Copper and silver are also found in the combined state as their sulphide or oxide ores. The oxides of these metals can be reduced to metals by heating alone.

EX :- When cinnabar (mercury sulphide) is heated in the presence of air, it is converted into mercury oxide. And when mercury oxide is heated, it gives mercury metal.

$$\underset{\substack{\text{Cinnabar}\\ \text{(mercury sulphide)}}}{2HgS\,(s)} + \underset{\text{Oxygen}}{3O_2} \xrightarrow{Heat(Roasting)} \underset{\text{Mercury (II) oxide}}{2HgO\,(s)} + \underset{\text{Sulphur dioxide}}{2SO_2(g)}$$

The mercury (II) oxide obtained after roasting is heated again strongly which turns mercury (II) oxide into mercury metal.

$$\underset{\text{Mercury(II) oxide}}{2HgO\,(s)} \xrightarrow{Heat\ (Reduction)} \underset{\text{Mercury}}{2Hg} + \underset{\text{Oxygen}}{O_2(g)}$$

(ii) Extracting Metals in the Middle of the Activity Series:

The metals in the middle of the activity series such as iron, zinc, lead, copper, etc., are moderately reactive. These are usually present as sulphides or carbonates in nature. The metal sulphides and carbonates are converted into metal oxides. As It is easier to obtain a metal from its oxide, as compared to its sulphides and carbonates.

Converting into oxides:

(a) Roasting : The sulphide ores are converted into oxides by heating strongly in the presence of **excess air**. This process is known as roasting.

Ex :- Zinc blende [ZnS (Zinc sulphide)] first converted into zinc oxide by the process of roasting. In the process of roasting, zinc blende [ZnS (Zinc sulphide)] is heated strongly in the presence of excess of air, i.e. oxygen, this converts zinc blende into zinc oxide.

$$2ZnS(s) + 3O\ (g) \xrightarrow{Heat} 2ZnO(s) + 2SO_2\ (g)$$

(b) Calcination: The carbonate ores are changed into oxides by heating strongly in **limited air**. This process is known as calcination.

Ex :- Calamine [$Zn(CO)_3$(Zinc carbonate)] is heated in the presence of limited air.

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$$ZnCO_3\,(s) \xrightarrow{Heat} ZnO(s) + CO_2\,(g)$$

(c) Reduction:- The process of converting metal oxides to respective metals by heating is called Reduction..The metal oxides are then reduced to the corresponding metals by using suitable reducing agents such as carbon.

Ex :-

$$ZnO(s) + C\,(s) \xrightarrow{Heat} Zn(s) + CO(g)$$

Reduction of metals oxide to respective metal by using reducing agent other than carbon (coke)

Besides using carbon (coke) to reduce metal oxides to metals, sometimes displacement reactions can also be used. The highly reactive metals such as sodium, calcium, aluminium, etc., are used as reducing agents because they can displace metals of lower reactivity from their compounds.

Ex :- when manganese dioxide is heated with aluminium powder, the following reaction takes place –

$$3MnO_2(s) + 4Al(s) \rightarrow 3Mn(l) + 2Al_2O_3(s) + Heat$$

Thermit Reaction: Some displacement reactions are highly exothermic. the amount of heat evolved is so large that the metals are produced in the molten state. such a reaction is known as thermit reaction.

Ex :- The reaction of iron(III) oxide (Fe_2O_3) with aluminium is used to join railway tracks or cracked machine parts.

$$Fe_2O_3(s) + 2Al(s) \rightarrow 2Fe(l) + Al_2O_3(s) + Heat$$

(iii) Extracting Metals towards the Top of the Activity Series:

The metals high up in the reactivity series are very reactive. These metals are obtained by electrolytic reduction. For example, sodium, magnesium and calcium are obtained by the electrolysis of their molten chlorides. The metals are deposited at the cathode (the negatively charged electrode), whereas, chlorine is liberated at the anode (the positively charged electrode).

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Ex :- When electric current is passed through molten sodium chloride, sodium metal is liberated at cathode and chlorine is liberated at anode.

$$Na^+ + e^- \rightarrow Na$$

$$2Cl^- \rightarrow Cl_2\,(g) + 2e^-$$

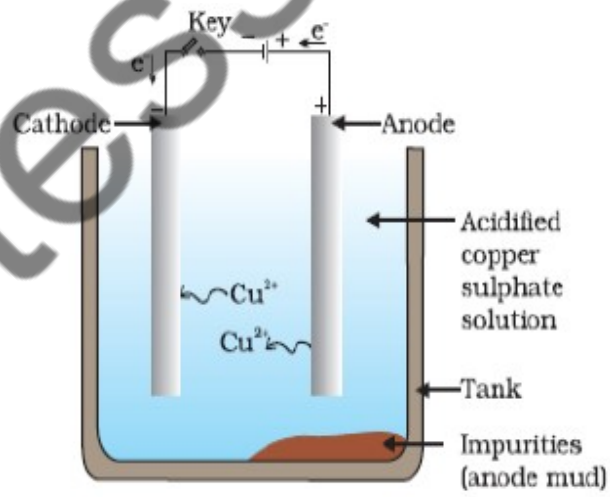
$$NaCl \xrightarrow{electric\ current} Na\,(s) + Cl_2(g)$$

Refining of Metals

The metals produced by various reduction processes described above are not very pure. They contain impurities, which must be removed to obtain pure metals. The most widely used method for refining impure metals is electrolytic refining.

Electrolytic Refining:

Many metals, such as copper, zinc, tin, nickel, silver, gold, etc., are refined electrolytically. In this process, the impure metal is made the anode and a thin strip of pure metal is made the cathode. A solution of the metal salt is used as an electrolyte. On passing the current through the electrolyte, the pure metal from the anode dissolves into the electrolyte. An equivalent amount of pure metal from the electrolyte is deposited on the cathode. The soluble impurities go into the solution, whereas, the insoluble impurities settle down at the bottom of the anode and are known as anode mud.



At anode $Cu - 2e^- \rightarrow Cu^{2+}$

At cathode $Cu^{2+} + 2e^- \rightarrow Cu$

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Corrosion

When some metals articles come in the contact of oxygen or moistures, this reacts with oxygen or water vapour and forms metal oxides which get rusted, resulting in fatal surface loss of metals. This surface loss of metal is known as corrsion.

Ex:-

- When iron metal comes in contact with oxygen present in moist air, a brown coat deposited over it because of formation of iron oxide. This is called **rusting of iron.**

$$\text{Iron} + \text{Water} + \text{Oxygen} \rightarrow \text{Rust } (Fe_2O_3n.H_2O)$$

- When silver metal comes in contact with sulphur present in air, a black coat is deposited over it because of formation of silver sulphide . This is called corrosion or tarnishing of silver.

- When copper comes in contact with carbon dioxide present in air, a green coat is deposited over it because of formation of copper carbonate. This is called corrosion or tarnishing of copper.

Prevention of Corrosion :- Corrosion can be prevented by

(i) By painting, Oiling and by greasing.

(ii) By Galvanization, chrome plating or by anodising.

(iii) By making alloys.

Galvanization : A method of protecting steel and iron from rusting by coating them with a thin layer of zinc. This process is known as Galvanization.

Alloying :- Pure metals are not used to make articles. So there are mixed some other substances to make them hard and strong and causing changing in metal's properties. This process is called alloying. . Alloying improves the qualities of metal, such as strength, resistance to corrosion, ductility, malleability, etc.

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Examples of alloy

1. Bronze :- Bronze is made by mixing 88% of copper and 12% of tin. Bronze is used in making of ship propellers, submerged bearings, statues, etc.

2. Brass :- Brass is the alloy of copper and zinc. Brass is used in making statues, musical instruments, etc.

3. Solder :- Lead is mixed with tin make Solder .

4. stainless steel :- Iron is mixed with nickel and chromium to make stainless steel.

5. Alloying of gold :- Gold is very soft metal because of which making of jewelry is difficult with pure gold. Thus, to increase its hardness, copper or silver is mixed with gold.

NOTE :- Pure gold is said to be of 24 carats. Gold jewelries available in market are usually of 22 carat. This means 22 carat of gold contains 2% of copper or silver and 98% of gold.

6. Amalgam :- In alloy if one of the metals is **mercury**, then alloy is called amalgam. Thus, amalgam is the homogeneous mixture of mercury and other metals.

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